

Sequence models for sound: decoding audio from EEG with LLM architectures

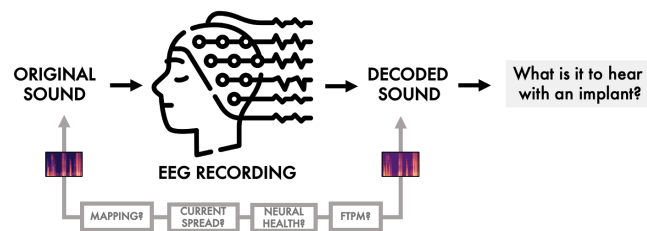
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Linked: [SPiN 2026 - reconstructing sound from EEG with LLMs](#) • [WCA 2026 - lexical surprise / codec EEG](#)

Can large-language-model architectures recover how the brain encodes sound?

Reconstructing sound from EEG with LLMs

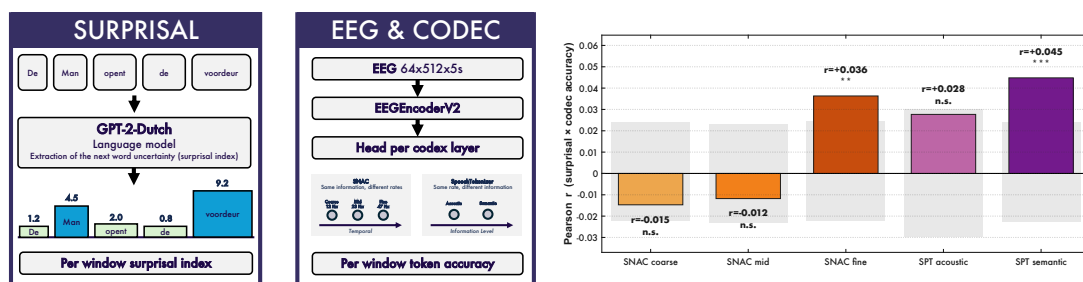
Existing decoders (linear envelope tracking, CNN/RNN spectrogram models) recover only specific acoustic components. I built a framework on large-language-model architectures - NeuroLM, a GPT-2-based model pre-trained on large-scale EEG - with sound tokenisation, to attempt full sound reconstruction from brain activity.



The reconstruction paradigm and its link to the individual factors studied.

Lexical surprise / codec EEG

Using an audio codec over 64-channel EEG and word-level surprisal from a language model, this study asks whether lexical surprise selectively amplifies phonetic versus semantic neural codes. Only the codec streams carrying lexical information correlated with the surprisal index.



Stages of the lexical-surprise study.